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1.0 OBJECTIVE

1.1 This SOP defines how the GasWorks software for the Prima BT Device is administered. Policies governing the Software's 21 CFR Part 11 compliance, roles and permissions, storage and maintenance of data, audit trail, and periodic inspection are described by this SOP.

2.0 SCOPE

2.1 This SOP applies to all employees who are responsible for administering the Software for the purposes of setting 21 CFR Part 11compliance, adding and removing users, and managing data, data backups and the audit trail.

3.0 DEFINITIONS

3.1 Administrator: User with the highest level of access control over the Software / System. The Administrator is also responsible for providing access to users, setting levels of access appropriate to job roles, disabling access for users who no longer require it, backing up system data, restoring system data, and setting the audit trail. If the Software / System has the capability to be FDA 21 CFR Part 11 Compliant, then the Administrator is responsible for enabling compliance.

3.2 I/O Cabinet: Input / Output Cabinet.

3.3 Local Control Software: Software embedded in equipment that enables direct, on-site monitoring and control of operations without needing a central system.

3.4 Supervisor: User who supervises general users (operators) of Software / System.

3.5 User: General user (operator) of the Software / System.

4.0 POLICY

4.1 Only trained System Administrators shall perform tasks within the scope of this document.

4.2 System Administrators shall have exclusive access to change the configuration settings of the GasWorks software.

4.3 Calibration and software updates shall be performed to ensure compliance with operational and quality standards.

4.4 System logs and data integrity shall be maintained as per 21 CFR Part 11 compliance.

5.0 RESPONSIBILITY

5.1 Administrator

5.1.1 There shall be two (2) Administrators assigned to the Prima BT Bench Top Process Mass Spectrometer: Assigning two Administrators ensures that a backup is available if one Administrator is unavailable. These Administrators should not have a direct stake in the data/reporting generated by the Unit. (The Administrators' stake in data is limited to the accuracy, accessibility, availability, and reliability of the reporting function).

5.2 Manufacturing Department

5.2.1 Performing system calibration through the software either once a month or per batch, whichever occurs first.
5.2.2 Changing the filament as prompted by the software.
5.2.3 Ensuring calibration is performed on the exact date provided by Maintenance. Any delay will result in an overdue entry in BMRAM.
5.2.4 Taking and managing reports of stream analysis.

5.3 Maintenance Department

5.3.1 Replacing the calibration gas cylinders when they are depleted.

5.4 Quality Assurance Validation

5.4.1 Quality Assurance is responsible for reviewing and approving all documentation related to the administration and use of the Gasworks Software for the Prima BT Bench Top Process Mass Spectrometer.

6.0 SYSTEM OVERVIEW

GasWorks is a comprehensive software platform developed by Thermo Fisher Scientific for the control and monitoring of process mass spectrometers. It is engineered to support complex gas analysis tasks, offering a robust set of tools for instrument configuration, automated scheduling, calibration routines, and real-time as well as historical data visualization. Display options include numeric tables, trend graphs, and bar charts, enhancing data interpretation. The software supports several measurement modes, such as Single Stream Analysis, Multiple Ion Monitoring (MIM), Histogram Scan, Analog Scan, and Leak Detection, catering to both standard and advanced analytical operations.

Operating on both a PC and the instrument's internal processor, GasWorks ensures synchronized control and data exchange across platforms. It features extensive customization for UI appearance, language, alarms, and communication protocols. Integrated security tools—like encrypted data handling, audit logs, and user access controls—support compliance with industry regulations. These elements make GasWorks V3 a powerful, flexible, and user-friendly solution for continuous gas monitoring in industrial environments.

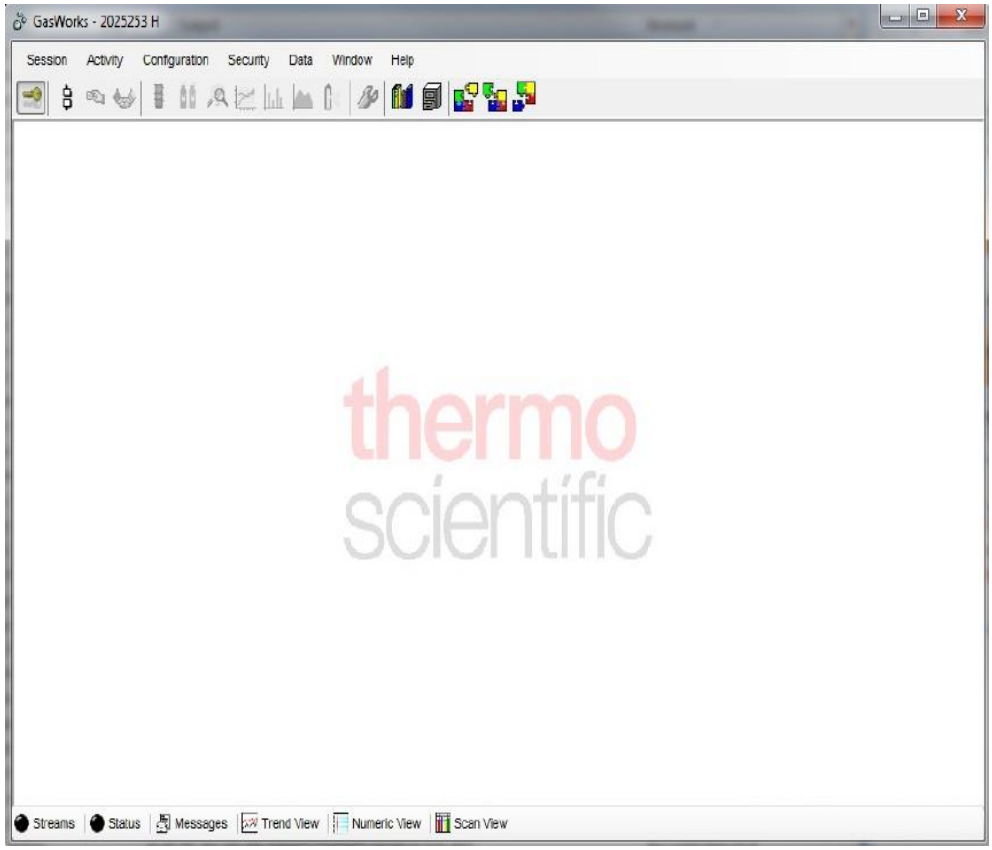

Figure 1: GasWorks Home screen

6.1 Key Features:

6.1.1 Multi-stream Sampling

Rapid Multi-Stream (RMS) capability enables fast analysis across multiple gas streams.

6.1.2 Automated Calibration

Scheduled calibration routines ensure consistent and accurate gas composition analysis.

6.1.3 Data Logging & Reporting

Real-time data logging and trend analysis for performance monitoring and documentation.

6.1.4 21 CFR Part 11 Compliance

Secure user authentication and full audit trails support regulatory compliance.

6.1.5 Communication Protocols

Supports Modbus, OPC, and Siemens 3964R for integration with various process control systems.

7.0 REFERENCES

7.1 Title 21 – Food and Drugs, Chapter 1 – Food and Drug Administration, Department of Health and Human Services, Subchapter A – General, Part 11 Electronic Records; Electronic Signatures (21 CFR Part 11)

7.2 SOP for Data Backup and Disaster Recovery (IP0037)

7.3 Computerized System Policies (IP0129)

8.0 PROCEDURES

8.1 Start-up and shutdown

8.1.1 Start-up Procedure

Before launching the GasWorks software, ensure the connected mass spectrometer instrument is correctly powered on and ready for operation. To start the GasWorks software:

8.1.1.1 Power on the PC, monitor, and printer (if applicable).

8.1.1.2 Launch the GasWorks application by selecting it from the Start Menu or by double-clicking the GasWorks icon on the desktop.

8.1.1.3 Establish a connection between the PC and the instrument:

- To connect manually, navigate to Activity → Connect from the main menu.
- To enable automatic connection on start-up, configure the setting under Session → Preferences.

Note: When GasWorks starts, it automatically checks the version of each configuration database. If a mismatch is detected between the database version and the software build (e.g., build 500), the software will generate an error and terminate. If this occurs, contact Thermo Fisher Scientific Technical Support for further assistance.

8.1.2 Shutdown procedure

The shutdown procedure for GasWorks depends on the current operating state of the system. Always ensure that data acquisition is safely stopped, and the instrument is properly disconnected before closing the application.

To shut down GasWorks:

8.1.2.1 Stop any ongoing analysis or measurement processes.

8.1.2.2 Disconnect the software from the instrument by selecting Activity > Disconnect.

8.1.2.3 Close the GasWorks application.

Note: If the instrument is configured for standalone operation, the analysis process may continue independently after disconnection. In such cases, it is safe to close the software without interrupting the ongoing analysis.

8.2 Setting Up the Administrator Account

8.2.1 Turn the instrument's power switch to the ON position. Once powered, the system will display the Home Screen.

Figure 2: Accessing Security

8.2.2 To access the security management program, select Security → Permissions from the GasWorks main menu.

8.2.3 If you have followed the instructions above, the GasWorks Security Login screen will appear, as shown in Figure 3.

Figure 3: GasWorks Security Login screen

8.2.4 Use default login credentials to access the unit for the first time. Enter the credentials as shown in Figure below and press the "OK" button to proceed.

Login Name	*root*
Password	*password*

Figure 4: Default Administrator

Note: The name is not case-sensitive. The password is case-sensitive. Once logged in, the Administrator should immediately change the default login credentials for security purposes.

8.3 Changing Administrator Login Credentials

8.3.1 To modify administrator details or to add or remove an administrative user, navigate to the menu bar in the security program and select: Security → Administrative Users This will open the Administrative Users window, as shown in the figure below.

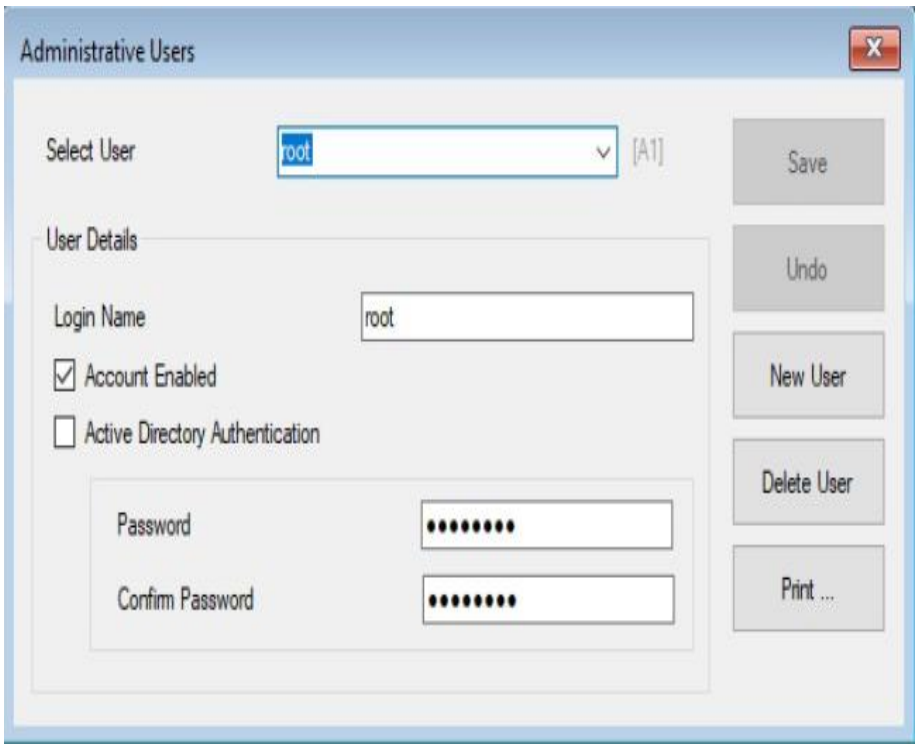

Figure 5: Administrative Users window

8.3.2 In the Administrative Users window, fill in all required user details, including the username and password fields. Once completed, click the Save button to apply the changes.

8.3.3 The following table provides a description of each field within the Administrative User Details section

Select User	**The full name for the user.**
Login Name	The login user name.
Account Enabled	By default, the account is not enabled. To enable the administrative users profile, check this box.
Active Directory Authentication	Check this box if the login name is to be authenticated via Active Directory.

Select User	The full name for the user.
Password	Password for this administrative user (not applicable for Active Directory).
Confirm Password	Confirmation of the password (not applicable for Active Directory).

8.3.4 User can add additional administrator accounts by clicking the **New User** button shown in the figure above and filling in the required details as described above.

Note: Make sure that you keep a careful note of the new settings. If you do forget the administrator's name or password, your only option is to contact your Thermo Fisher Scientific representative, who may be able to assist. It is not possible for an administrative user to delete their own account details.

8.4 User Management

8.4.1 Setting Up New Groups

8.4.1.1 To create and configure new user groups with specific access levels, open the Group Details window by selecting Edit → Group Details from the menu bar or by clicking the Group Details button on the toolbar.

8.4.1.2 In the Group Details window, as shown below, select the option to create a new group.

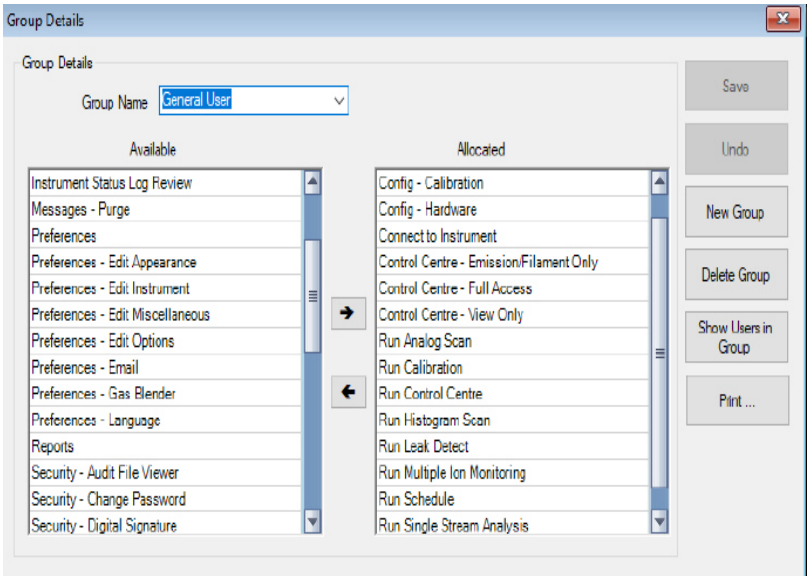

Figure 6: Group Details window

8.4.1.3 Enter a name for the new group.

8.4.1.4 Assign security functions to the group:

8.4.1.5 Use the right arrow (→) to move desired functions from the Available column to the Allocated column.

8.4.1.6 Use the left arrow (←) to remove functions from the Allocated column if needed.

8.4.1.7 To verify or manage users within the group, click the Show Users in Group button.

Note: *"Internal Use" and "GasWorks Host" are predefined Groups. The "Internal Use" group functions correspond to a "No User" state, that is, when there is no logged-in user. This is the state in which GasWorks starts up when launched, and no one has yet logged in.*

8.4.2 Setting Up New Users

8.4.2.1 The GasWorks security window allows you to create new users or modify the details of existing users. To access this window, either:

8.4.2.2 From the main window, go to Security → Permissions, then enter the administrator password in the GasWorks Security Login window.

Figure 7: GasWorks Security Login window.

8.4.2.3 Once logged in, the GasWorks Security window opens—select Single User to add or modify users.

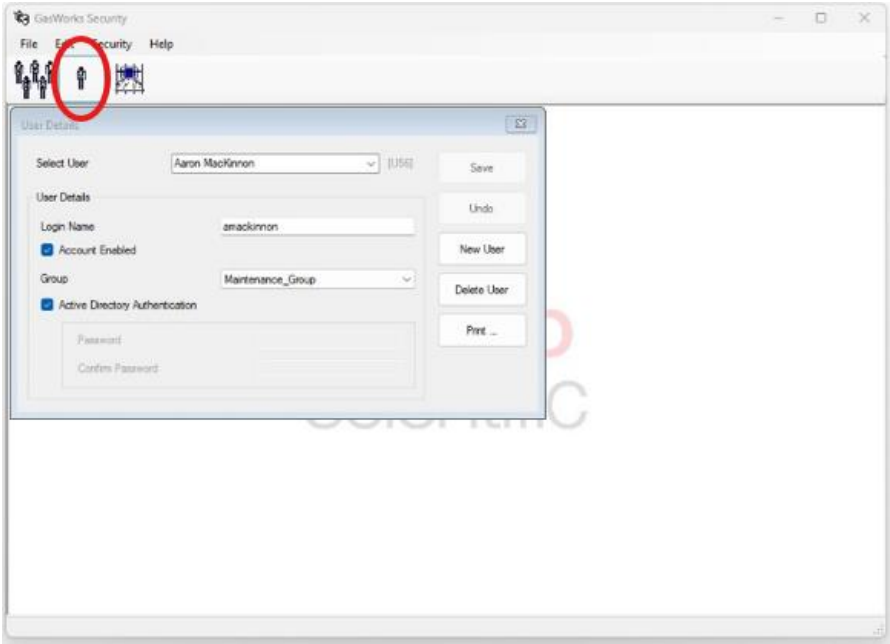

Figure 8: GasWorks Security window

8.4.2.4 For this system, Active Directory Authentication is used. To set up a new user, select the user from the directory, enter the BioVectra HMI login name as the Username, select Account Enable, choose the appropriate Group, check Active Directory Authentication, and click New User to create the user (see figure below).

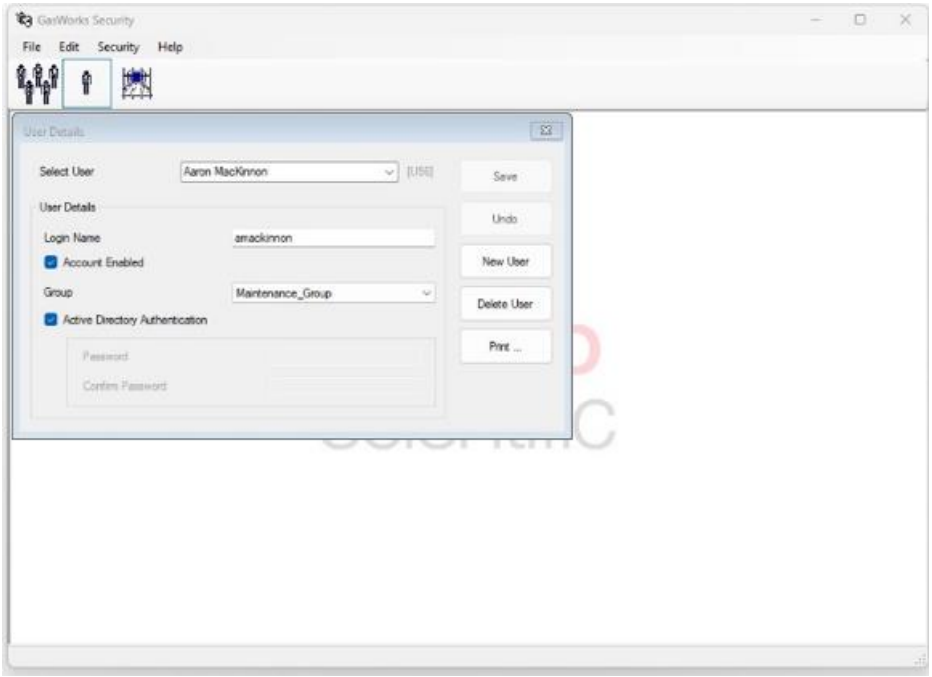

Figure 9: User Details window with Active Directory Authentication enabled and new user details entered.

8.4.2.5 The following table provides a description of each parameter in the User Details window:

Parameter	Description
Select User	This is the full name for the user.
User Name	This is the login name for the user.
Account Enabled	By default, the account is not enabled. To enable the users profile check this box.
Group	Select the Group that the user is to be a member of from the drop down list.
Active Directory Authentication	Check this box if the login name is to be authenticated via Active Directory.

8.5 System Configurations

8.5.1 The System Options window is used to configure all security-related settings within GasWorks as shown in the figure below. To access this window, either:

8.5.2 Select Edit → System Options from the menu bar in the security program, or

8.5.3 Click the System Options button on the toolbar.

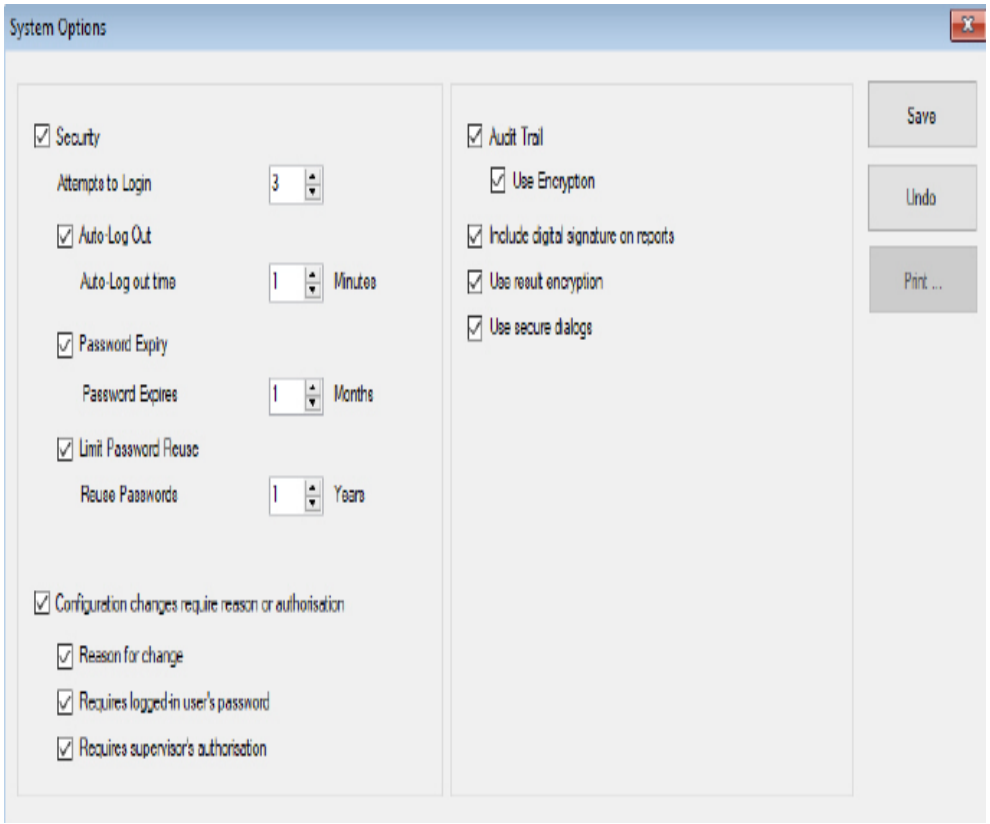

Figure 10: System Options window

8.5.4 Once opened, the **System Options** window allows you to configure parameters as described below:

Parameter	Description
Security	Enables GasWorks security features. All indented security controls only take effect if this box is checked.
Attempts to Login	Sets the number of failed login attempts before a user is locked out. To reinstate the user, uncheck and re-check the Account Enabled box in the User Details window.
Auto-Log Out	When enabled, it automatically logs out inactive users after a set number of minutes. Applies even if security is disabled.
Password Expiry	Forces users to reset their password after a specified time (in months).
Limit Password Reuse	Prevents users from reusing passwords within a defined time period (in years).
Configuration Changes Require Reason or Authentication	Defines validation requirements for saving changes:
Reason for Change: Requires entry of a reason before applying configuration changes. Audit trail must be enabled.	
Requires Logged-in User's Password: Prompts for the current user's password before changes are applied. Security must be enabled.	
Requires Supervisor's Authorization: Requires a supervisor's login credentials to authorize the configuration change. Security must be enabled.	
Audit Trail	Creates a log file (CSV format) of all configuration changes with time/date stamps. Optionally encrypts logs in audit format.
Include Digital Signature on Reports	Adds a digital signature to Calibration, Statistical, and Alarm reports. Requires a valid certificate setup.
Use Result Encryption	Encrypts all analysis results, making them viewable only through the GasWorks Encrypted Results Viewer.
Use Secure Dialogues	Disables file renaming, deletion, and drag-and-drop actions by replacing Windows File Explorer with secure dialogues for file operations.

8.6 Backing Up or Restoring Data

This section provides comprehensive guidance for backing up and restoring the GasWorks software. It is the user's responsibility to implement and verify backup and recovery procedures and—most importantly—to regularly test them to ensure effectiveness.
All backup and restoration files shall be stored in a secure, access-controlled location designated by the system administrator, such as an approved network share or external storage device compliant with site data retention policies. For BioVectra, the designated storage location is BioVectra-SRV-Backup located at D:\Data Backup.

Note: *This section only covers the backup and recovery of the main GasWorks software. If you are using additional components from the GasWorks distribution (e.g., Wizards, utilities like OPC, manuals), further backup strategies may be needed.*

8.6.1 Backup

8.6.1.1 After the installation and configuration of the GasWorks software package, the following initial backup tasks should be performed:

8.6.1.2 Record the GasWorks version and build number (accessible via Help → About in the application).

8.6.1.3 Archive the installation media or retain a secure copy.

8.6.1.4 Create a file-level backup of the GasWorks installation directory (typically C:\GasWorks) onto suitable backup media.

8.6.1.5 This initial backup captures a clean, working state of the software and is essential for full system recovery in the event of hardware failure or accidental file loss.

Important: Both the GasWorks installation media and a full backup of the GasWorks directory are required for a complete system recovery.

Note: These same steps can also be used to move a GasWorks installation from an existing PC to a new one (via USB or network).

8.6.1.6 Complete Local Backup

Maintaining a local backup copy of the GasWorks installation folder helps safeguard against accidental file deletion.

Steps:

8.6.1.6.1 Check Available Disk Space
Use folder properties (right-click → *Properties*) to ensure sufficient disk space for the backup.

8.6.1.6.2 Identify the Installed Location
In GasWorks, navigate to Help → About. The typical installation path is: C:\GasWorks\GasWorks.exe

8.6.1.6.3 Stop the GasWorks Application

8.6.1.6.4 Create a Folder Copy
Use File Explorer to copy the entire C:\GasWorks folder to another location or device.

8.6.1.6.5 Archive to Backup Media
Save this backup to an external storage medium (e.g., external drive, network share).

Note: The data folder (typically C:\GasStore) may contain large data files. Back up this folder as needed.

8.6.1.7 Backup of Salient Files Only

8.6.1.7.1 In cases where a full backup isn't feasible, back up at least the critical and important files within the GasWorks directory:

8.6.1.7.2 Critical Files (must be backed up):

8.6.1.7.3 The following table outlines the files that must be backed up.

File	Description
C:\GasWorks\DB\instconf.accdb	Instrument configuration settings (essential for recovery)

8.6.1.7.4 Important Files (recommended for backup):

File	Description
C:\GasWorks\DB\security.accdb	Required if user security is enabled
C:\GasWorks\DB\trends.accdb	Contains trend display configuration
C:\GasWorks\DB\messages.accdb	Stores alarm, warning, and status messages
C:\GasWorks\DB\rs.accdb	Contains user-generated result data
C:\GasWorks\gasworkssettings.xml	Stores configuration options

8.6.1.7.5 Also consider backing up custom configuration files such as alarm files, and:

8.6.1.7.6 Optional Report Directories:

- Alarmdet – CSV alarm reports
- Audit – Security audit trail

- Calrpt – Calibration reports (check Calibration Configuration for path)
- Messages – Instrument message logs
- Statsrpt – Statistical reports (check Analysis Configuration for path)

8.6.1.7.7 Data Storage Directory: C:\GasStore – May contain large data files; backup as required.

8.6.2 Recovery

Depending on your backup type, you can recover GasWorks using either a complete folder backup or important files only.

8.6.2.1 Recovery from Complete Backup

8.6.2.1.1 Copy the backed-up GasWorks folder to the same directory path on the recovery PC (usually C:\GasWorks).

8.6.2.1.2 Reboot the recovery PC and ensure no other applications are running.

8.6.2.1.3 Insert the GasWorks installation media.

8.6.2.1.4 If autorun doesn't start, manually run autorun.exe.

8.6.2.1.5 Select the appropriate instrument type and double-click the corresponding instrument icon.

8.6.2.1.6 Ensure the installation path matches the original (e.g., C:\GasWorks).

8.6.2.1.7 Reboot the system if prompted.

8.6.2.1.8 Launch GasWorks, hold the <F2> key and click Connect until you see either:

- "Clearing Database"
- "Updating Configuration"

8.6.2.2 Recovery from Important Files Only

8.6.2.2.1 Install GasWorks as described above (Steps 2–6 from the previous section).

8.6.2.2.2 Copy the previously backed-up critical and important files to their respective folders in C:\GasWorks\DB.

8.6.2.2.3 Launch GasWorks, holding down the <F2> key and clicking Connect, as described earlier.

8.6.2.3 Recovery to a Higher GasWorks Build (Upgrade Scenario)

Important: Before upgrading, check whether your instrument supports software upgrades or requires hardware ROM changes.

Steps:

8.6.2.3.1 Backup the following configuration databases from C:\GasWorks\DB:

- instconf.accdb (critical)
- security.accdb
- trends.accdb
- messages.accdb

8.6.2.3.2 Do not copy the following databases into a newer version: applicat.accdb/dbUpdate.accdb/engconf.accdb/language.accdb/langua APAC.accdb/rs.accdb

8.6.2.3.3 Backup any protocol or communication config files:

modmap.csv/opc-mb.csv/opc.ini (if applicable)

8.6.2.3.4 Install the latest version of GasWorks.

8.6.2.3.5 Copy the saved files into the corresponding C:\GasWorks and C:\GasWorks\DB locations.
Ensure no files are marked *read-only*.

8.6.2.3.6 Run GWUpdate.exe from C:\GasWorks to update database formats as needed.

8.6.2.3.7 If the update fails, contact Thermo Fisher Scientific Technical Support. Include:

- Zipped folders C:\GasWorks\DB and DBArchive
- Any log/error files (*.log, *.err) from C:\GasWorks
- Email to: techsupport.process.uk@thermofisher.com

8.6.2.3.8 Run IdentifyGW.exe from C:\GasWorks.

8.6.2.3.9 Confirm that the Build Version and Database Versions match the expected version (e.g., Build 500).

8.6.2.3.10 Reinstall supporting applications like VGOPCServer if required.

8.6.2.3.11 GasWorks is now ready for use.

8.7 Audit trails:

The Audit File Viewer is a separate application within the GasWorks software suite that allows users to review and analyze audit trail files generated by the system. This tool is accessible through the Security menu in GasWorks.

8.7.1 To open the Audit File Viewer:

8.7.1.1 Launch GasWorks.

8.7.1.2 Navigate to Security → View Audit Files.

8.7.1.3 When the Audit File Viewer launches, the main window appears blank initially.

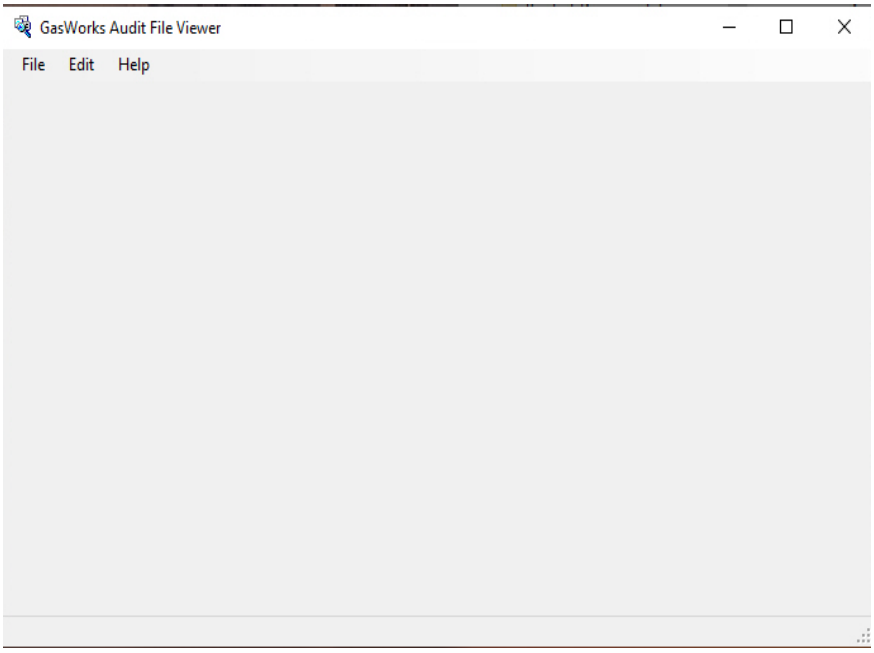

Figure 10: Audit File Viewer Window

8.7.2 To view audit trail files:

8.7.2.1 Select File → Open from the menu bar.

8.7.2.2 Navigate to the location where the audit trail files are stored.

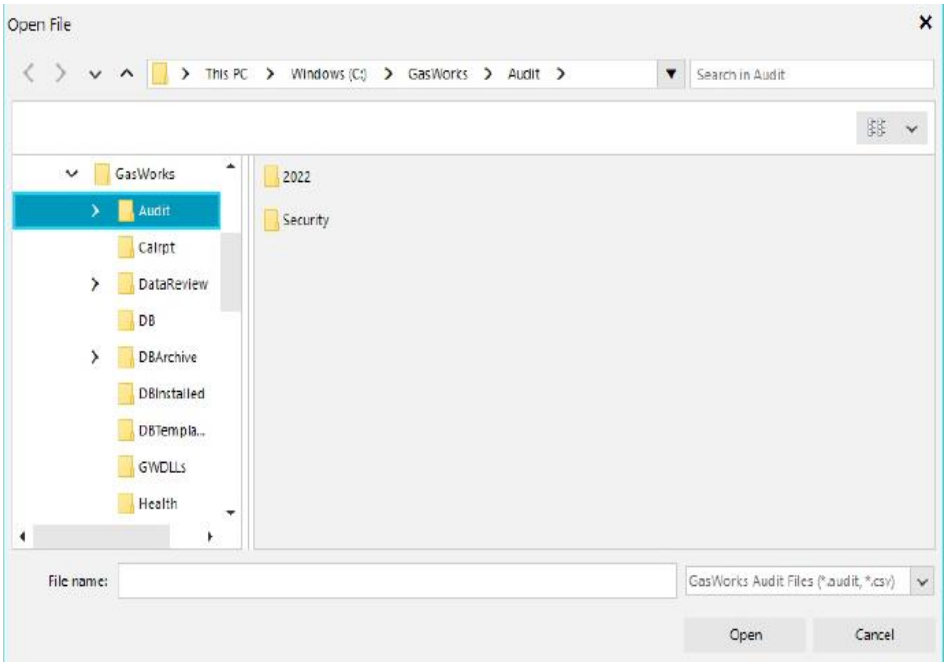

Figure 11: Open File window Path to audit trails

8.7.2.3 Select the desired file and click **Open**

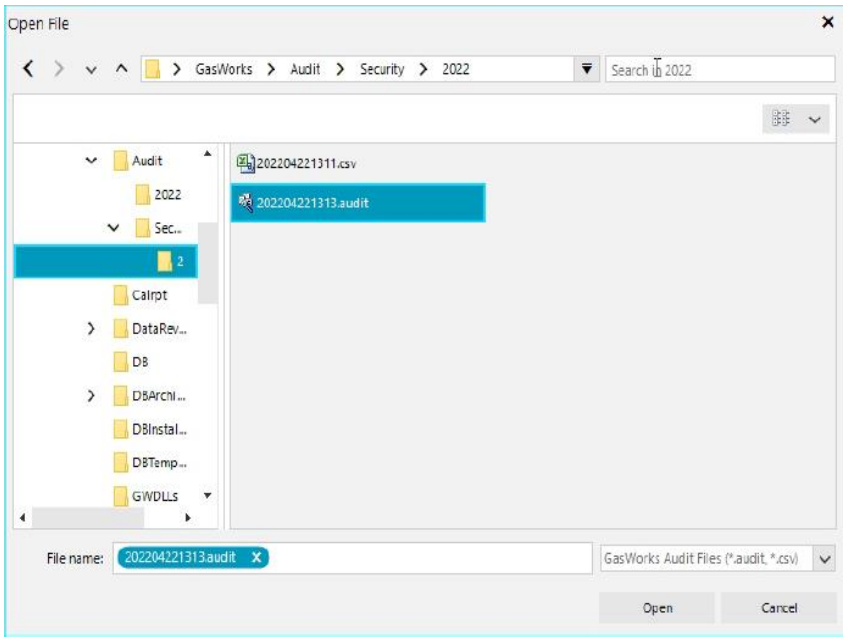

Figure 12: Open File window Path to audit trails

8.7.2.4 If the selected file is in **.csv** format, it will be displayed in plain text format directly in the viewer.

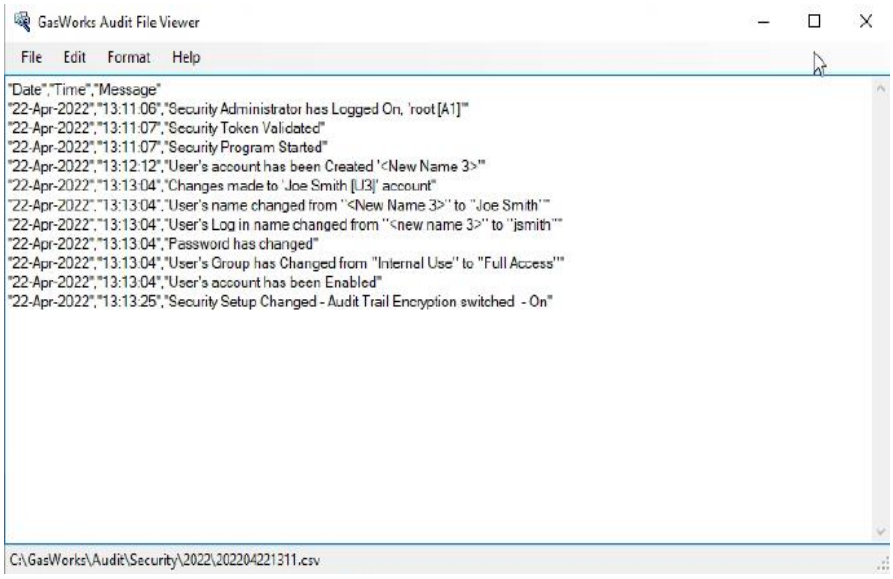

Figure 13: Plain text view of a .csv file displayed directly in the viewer

8.7.2.5 If the file is in .csv format the window will display the selected audit file in plain text.

8.7.2.6 If the selected file is in. audit (encrypted) format, additional metadata will be visible, including information about the file and the version of GasWorks that generated it.

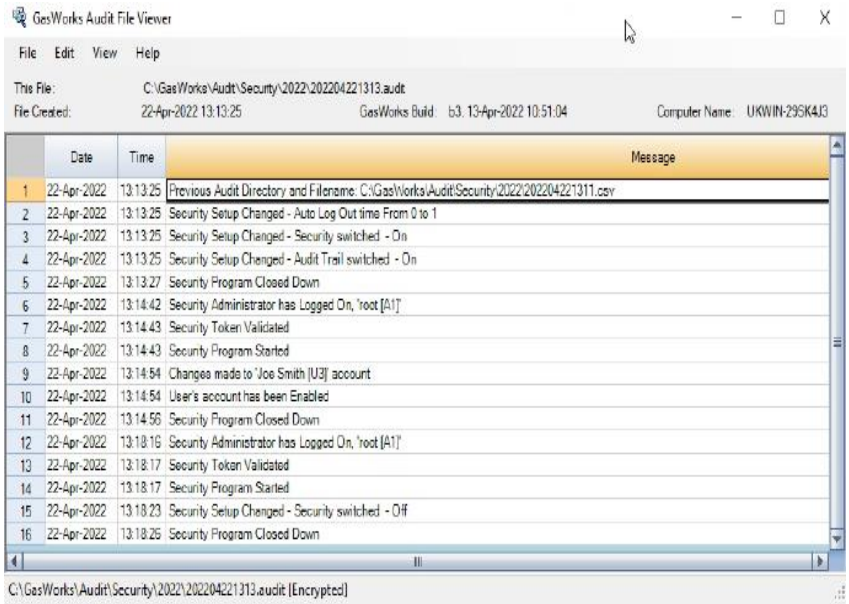

Figure 14: Display of additional metadata for an encrypted audit file

8.7.2.7 The encrypted format files also provide more information on the data which goes between the GasWorks program and the instrument. These are known as MASIF records.

8.7.2.8 MASIF Record Types

8.7.2.8.1 The encrypted **audit** files include **MASIF records**, which provide detailed logs of communication and configuration changes between the GasWorks software and the instrument. MASIF records are categorized into three types:

Record Type	Description	Color Displayed
MASIF User	Logs configuration data that may be modified through the GasWorks application.	Blue
MASIF Hardware	Logs details about the instrument hardware. These records remain unchanged after installation.	Green
MASIF Reprogramming	Logs software version changes when a new instrument firmware is installed.	Orange

8.7.2.8.2 By default, only MASIF User records are displayed in the viewer. To view other record types:

8.7.2.8.3 Go to the View menu and select or deselect the record types to display.

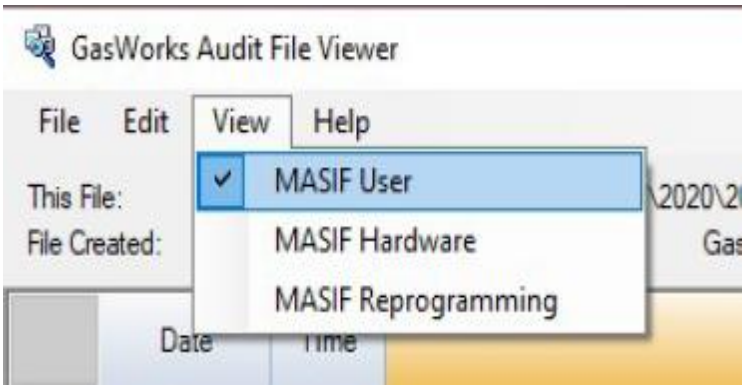

Figure 15: Default view showing only MASIF User records

Example Screenshots:

8.7.2.8.4 An example of MASIF User records.

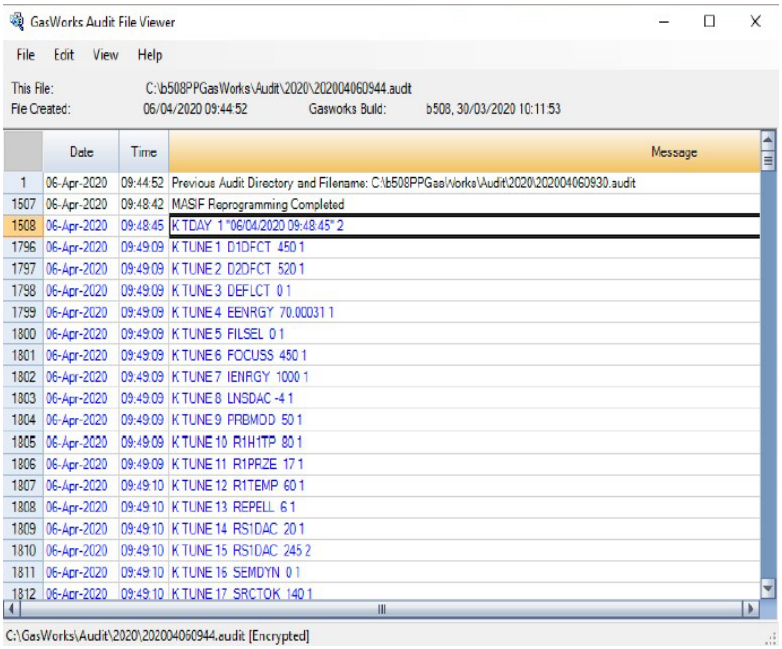

Figure 16: Example of MASIF User records

8.7.2.8.5 An example of MASIF Hardware records.

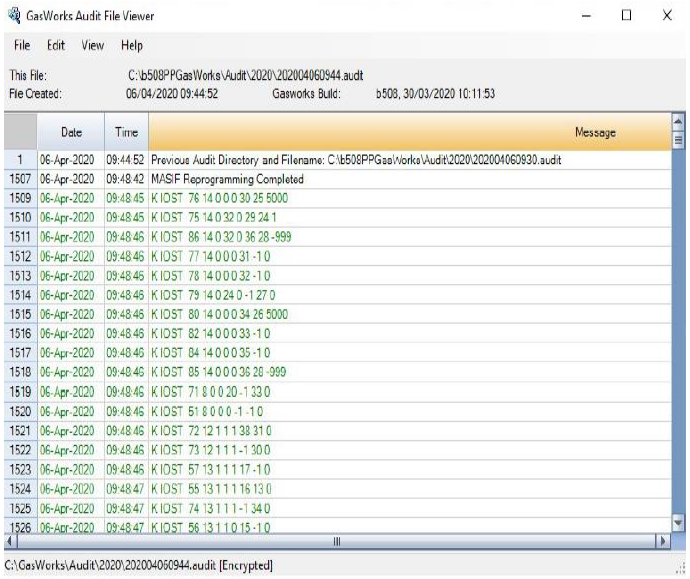

Figure 17: Example of MASIF Hardware records

8.7.2.8.6 An example of MASIF Reprogramming records.

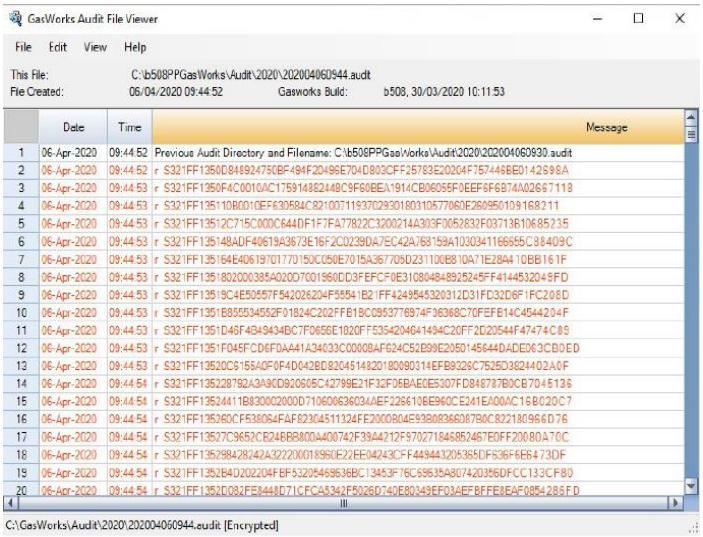

Figure 18: Example of MASIF Reprogramming records

8.7.3 Viewing Record Details

8.7.3.1 For MASIF User records, detailed field information is available:

8.7.3.2 Double-click any record entry in the grid.

8.7.3.3 A new window opens showing an expanded view of the selected record with field-level explanations.

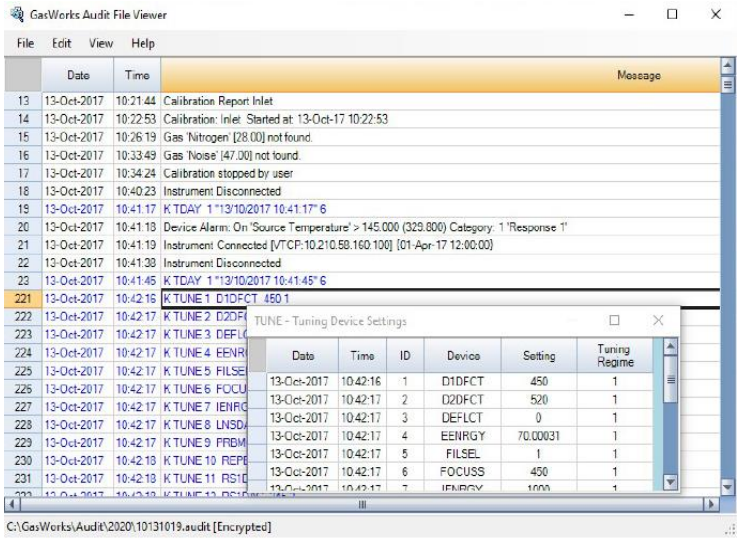

Figure 19: MASIF User records

8.7.4 Exporting Audit Data

8.7.4.1 The entire audit file, including the currently displayed MASIF records, can be exported to an Excel spreadsheet for reporting or archiving purposes.

8.7.4.2 To export:

8.7.4.2.1 Go to File → Save to Excel...

8.7.4.2.2 Choose a destination and file name for the exported spreadsheet.

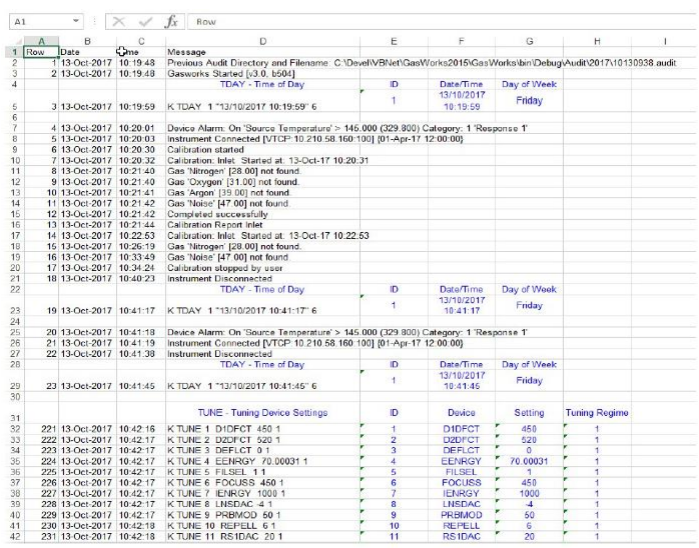

Figure 20: Exported Excel spreadsheets

9.0 TRAINING

9.1 All personnel who are designated as Administrators of the GasWorks Software must be trained in this procedure prior to use.

9.2 Training is accomplished by self-study

10.0 REVISION HISTORY

Revision 0, PKT24831, Request Date 10 Feb 2025
New Document